Predicting Compressive Strength and Optimal Replacement Level for Rice-Husk-Ash Blended Concrete from a Sparse Literature Dataset

September 13, 2026

Abstract

We build a machine-learning pipeline to predict the compressive strength and the optimal cement-replacement level of rice-husk-ash (RHA) blended concrete, using a dataset compiled from **79 research papers**. The central obstacle is data sparsity: strength values are packed inside free-text cells as *(replacement level × curing age)* grids. Parsing and exploding these cells—rather than any modeling choice—is the highest-leverage step, turning ∼70 packed rows into **756 clean data points across 44 papers**. Through a five-axis ablation study we find that (i) absolute strength is essentially unpredictable across papers ($R^2 < 0$) because the corpus mixes mortars, normal concrete and ultra-high-performance concrete, but becomes learnable once the paper's own control (0%) strength is supplied as an input; (ii) a strongly-regularized, monotonically-constrained gradient-boosted model on just three inputs *(replacement, age, control strength)* is the best predictor, reaching $R^2 = 0.50 \pm 0.20$**, MAE** $\approx$ **10 MPa**; and (iii) text embeddings (MiniLM/BERT-style) provide *no reliable gain* at this sample size. The optimal-replacement target cannot yet beat a naive prior—the derived predictor in fact degenerates to a constant. We conclude that the binding constraint is data quantity and the absence of base-mix descriptors, not model capacity.

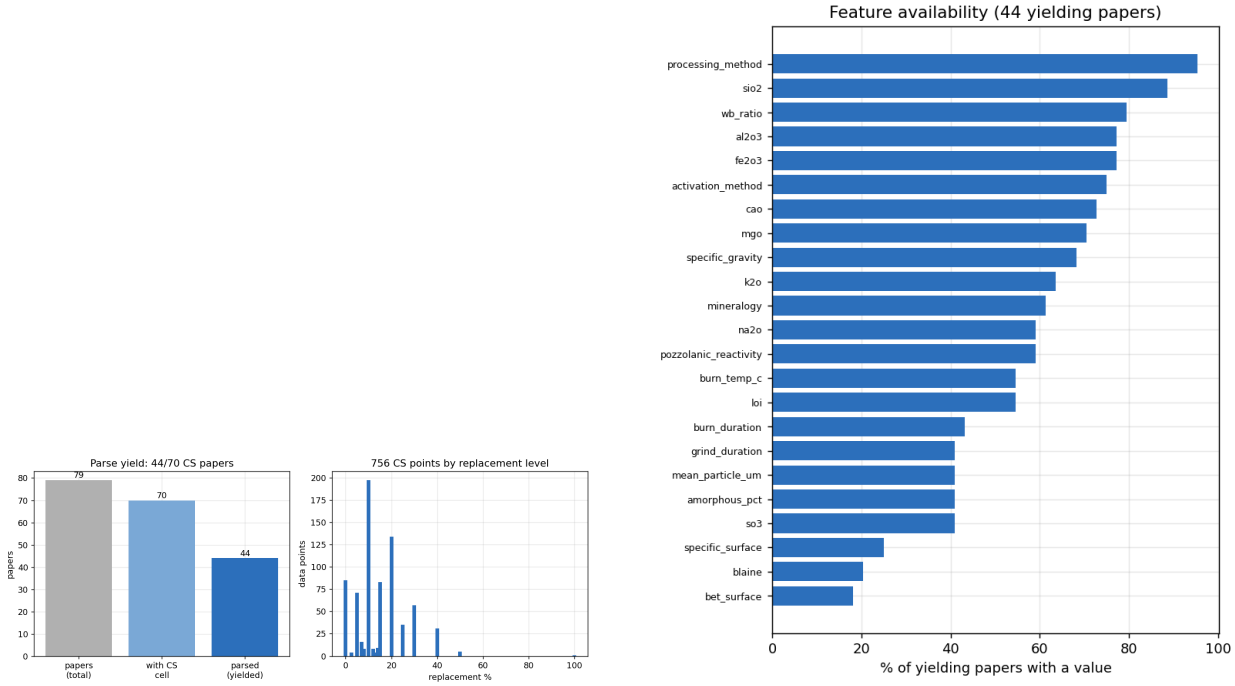
1 Introduction

Rice husk ash is a pozzolanic by-product that can partially replace Portland cement, reducing both cost and CO_2 footprint. The two engineering quantities of interest are the **compressive strength** of the resulting concrete and the **optimal replacement level**—the percentage of cement that can be replaced by RHA while maximizing strength. Our goal is a predictive model for both, trained on a dataset hand/LLM-extracted from the literature. The practical difficulties are (a) pervasive missing values and (b) selecting a model that performs well on a small, heterogeneous dataset. We take XGBoost as the baseline and interrogate the data and model choices rigorously.

2 The Dataset: What It Actually Contains

The source file `RHA Master sheet v2.csv` contains 225 columns organized under a **four-row hierarchical header**: category (→ PAPER METADATA/INPUT/OUTPUT), section, feature (55 features), and a sub-field. Every feature is split into four columns—`Extracted Value`, `Unit`, `Source Type`, `Exact Evidence`—of which only the first two carry predictive information; the latter two are provenance and are discarded. The file is `latin-1` encoded. The dataset contains **79 papers** with unique

titles. Sparsity is uneven rather than uniform: among the 44 papers that yield strength data, availability of the 23 retained features ranges from 96% (processing method) and 89% (SiO_2) down to 18% (BET surface), i.e. **5–82% missing**, with 8 of the 23 features more than half missing (Figure 1b). The prediction targets are present in 70 (strength), 71 (optimum), and 76 (replacement levels) papers.



(a) Parse yield and the distribution of recovered strength points by replacement level.

(b) Per-feature availability among the 44 yielding papers.

Figure 1: The dataset is small and sparse. Chemistry (SiO_2, oxides) and w/b are reasonably populated; fineness measures (BET, Blaine, specific surface) are scarce.

3 Data Representation: Parsing the Packed Cells

The single most valuable observation is that each Compressive Strength cell encodes a *grid* of measurements in one of ∼12 textual layouts, e.g.

```
3 days:  20 (7%), 20 (10%); 28 days:  44.5 (7%), 46 (10%); ...
```

We built a state-machine parser (`02_parse_explode.py`) that resolves, for each value, a *curing age* and a *replacement level* from inline qualifiers, segment headers, or persistent line context, and emits a tidy long table with one row per *(paper, replacement %, age)*. Precision safeguards include:

- unsigned number matching, fixing mis-reads such as `RHA8-10%` → 10 (not −10);
- a **per-paper whitelist**: a resolved replacement % must match the paper's declared levels (plus 0 for control), which rejects mis-labelled sample IDs (`RHA1`, `M2`) and recovers trailing-code mixes (`Aa-10`, `A35-10`);
- physical sanity clamps ($0 \le \% \le 100$, $0 < \text{age} \le 400$ d, $0.5 \le \text{MPa} \le 250$).

This recovers **756 data points across 44 of the 70 strength papers** (226 at the standard 28-day age). The 26 non-yielding papers are graph-derived narratives, pure ranges, or bare number lists with no age/level mapping—genuinely unrecoverable without the source PDFs, and logged to `parse_failures.csv`. Figure 2 checks the parsed data against expected physics, but both checks must be made *within* a paper: a raw cross-paper average is dominated by mix type (the corpus spans 1–205 MPa), not by the variable of interest. Normalising each *(paper, replacement level)* series by its own 28-day value, strength does rise with curing age—91% of pre-28-day points fall below the series' own 28-day value and 82% of later points exceed it, with the median ratio climbing from 0.53 at 1 d to 1.00 at 28 d and 1.21 at 90 d. The strength-vs-replacement shape, by contrast, is *not* reliably unimodal: of the 33 papers reporting at least three distinct replacement levels at 28 d, only 14 (42%) peak at an interior level, while 12 (36%) are highest at the lowest level tested—often the 0% control, meaning RHA never helped in that paper—and 7 (21%) are still rising at the highest level tested. An interior optimum is therefore common but far from universal, which is exactly why the optimum-prediction task in Section 8 is hard.

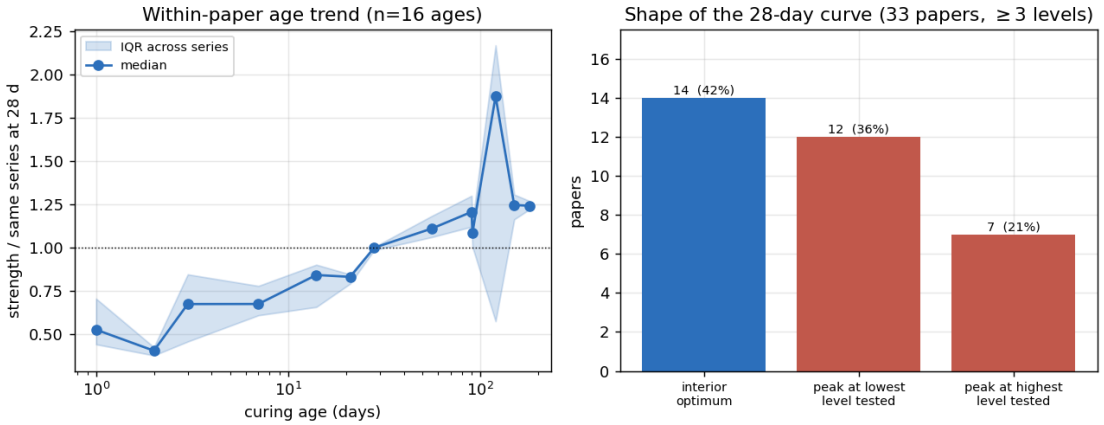


Figure 2: Physics checks on the parsed data, computed within papers. **Left:** each *(paper, replacement level)* series is normalised by its own 28-day strength; the median rises with age (shaded band is the IQR across series). Wiggles at 2, 91 and 120 d come from ages with very few series and are small-n noise, not a reversal of the trend. **Right:** the 28-day strength-vs-replacement curve has an interior optimum in only 42% of papers; in 36% the best level tested is the lowest one.

4 Feature Engineering

Numeric features (oxides, fineness, w/b, burn temperature/duration) are parsed as the mean of the numbers in each cell, robustly handling multi-sample entries such as "94.6 (RHA-A), 93.7 (RHA-B)". High-cardinality but semantically redundant text fields are **canonicalized** into compact categoricals: activation method ∈ {none, thermal, mechanical, both}; mineralogy ∈ {amorphous, mixed, crystalline}; processing ∈ {controlled, uncontrolled, . . . }; pozzolanic reactivity as an ordinal. We deliberately apply **no imputation** for the tree models, which handle missing values natively by learning a default split direction; imputers (median/KNN/MICE) are evaluated separately and only ever fit *inside* each cross-validation split to prevent leakage.

5 Methodology

Because rows from the same paper are strongly correlated, all evaluation uses **grouped cross-validation by paper**: no paper appears in both training and test. Given the small sample, a single split is unreliable, so we report the mean and standard deviation of **repeated grouped hold-out** (`GroupShuffleSplit`, 30% of papers held out, 40 repeats). We compare three target framings:

A. Absolute strength, all papers predict MPa directly from RHA features.

B. Relative-to-control predict strength normalized by the paper's 0% control.

C. Absolute strength given control predict the RHA-blend strength using the paper's control (0%) strength as an additional input—always known in practice, hence not leakage. Control rows themselves are excluded from evaluation to avoid a trivial identity.

6 Ablation Study

We ablate five axes. All numbers are repeated grouped hold-out (mean $\pm$ std) over 40 repeats, except the HistGB row of Study 2 and the three imputer rows of Study 4, where 31 repeats survived (a split is discarded when a held-out fold ends up with no usable rows). Comparisons *within* a study are like-for-like unless noted.

Study 1 — Target framing. Absolute strength across papers is *unpredictable* ($R^2 = -0.78$); the corpus mixes mortars, normal concrete and UHPC (1–205 MPa) whose baseline levels are set by mix factors absent from the data. Supplying the control strength (framing C) makes the RHA effect learnable (Table 1).

Table 1: Study 1: target framing (XGBoost, all structured features).

Framing	R^2	std	RMSE	MAE
C: blend | control strength	0.395	0.259	16.45	10.64
B: relative-to-control	−0.159	0.300	0.29	0.22
A: absolute, all papers	−0.776	1.549	30.99	20.95

Study 2 — Model. Random Forest and XGBoost dominate; HistGradientBoosting is unstable and linear Ridge fails badly on this tiny, collinear, missing-laden matrix (Table 2). Random Forest posts the higher mean here (0.436 vs 0.395), but the gap is roughly one-sixth of the run-to-run spread of either model ($\sigma \approx 0.21$–0.26) and so is not a reliable ordering. We nonetheless carry **XGBoost** forward, because the largest single gain in the whole study comes from the monotone constraints of Study 5 ($0.442 \rightarrow 0.482$), which XGBoost supports natively and `RandomForestRegressor` does not. The tuned monotone XGBoost of Section 7 reaches 0.50 ± 0.20, above the unconstrained Random Forest.

Study 3 — Feature set. Parsimony wins decisively: the three-feature *base* {replacement, age, control} is the best set, and *every* added feature group degrades generalization—clear evidence of overfitting at $n = 23$ papers (Table 3, Figure 3).

Table 2: Study 2: model (framing C, all features).

Model	R^2	RMSE
Random Forest	0.436	15.82
XGBoost	0.395	16.45
Mean baseline	−0.310	23.97
HistGB	−0.524	22.28
Ridge	−14.7	44.74

Table 3: Study 3: feature set (framing C, XGBoost).

Feature set	R^2	RMSE
base [repl, age, control]	0.442	15.54
base + fineness	0.435	15.73
base + categorical	0.427	15.93
all + MiniLM text	0.426	16.13
all + TF-IDF text	0.403	16.42
all structured	0.395	16.45
base + all numeric	0.349	16.85
base + chemistry	0.317	17.18
base + processing	0.236	17.69

Study 4 — Imputation. For the tree models, native NaN handling matches explicit imputation on R^2: native, median and KNN sit within 0.007 of one another, and only MICE clearly hurts (Table 4). The lower RMSE of the median and KNN rows is *not* evidence that they generalize better—those rows completed 31 of the 40 repeats, so their error is averaged over an easier subset of splits and is not directly comparable to the native row's 40. Since no imputer improves R^2 and each adds a fitted component that must be re-fit inside every split, we keep **native handling**.

Study 5 — Physics priors. Adding **monotone constraints**—strength non-decreasing in curing age and in control strength—gives the single best result, $R^2 = 0.482 \pm 0.217$, encoding domain knowledge the tiny sample cannot reliably learn on its own (Table 5).

Table 4: Study 4: imputation (framing C).

Scheme	R^2	RMSE
XGB + native	0.397	16.40
XGB + median†	0.396	14.88
XGB + KNN†	0.390	14.92
XGB + MICE†	0.249	16.35

†31 of 40 repeats; RMSE not comparable to the native row.

Table 5: Study 5: physics priors (base features).

Prior	R^2	RMSE
monotone(age, control)	0.482	15.02
log + monotone	0.458	15.71
plain	0.442	15.54
log-target	0.423	16.09

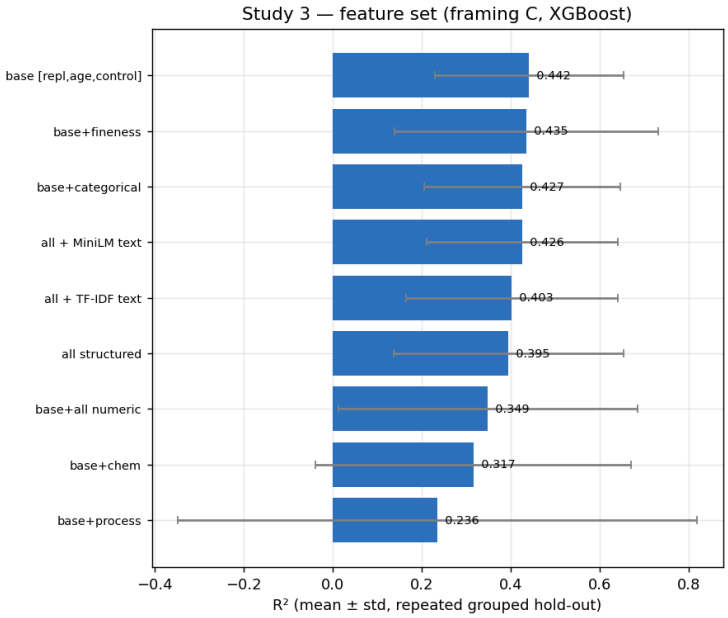


Figure 3: Feature-set ablation: parsimony wins (repeated grouped hold-out, mean ± std).

7 Final Model and Results

Features used (framing C). The final model takes exactly three inputs—no RHA chemistry, physical, or processing features are included, and Study 3 above is the evidence for why (every added feature group made generalization *worse* at $n = 23$ papers):

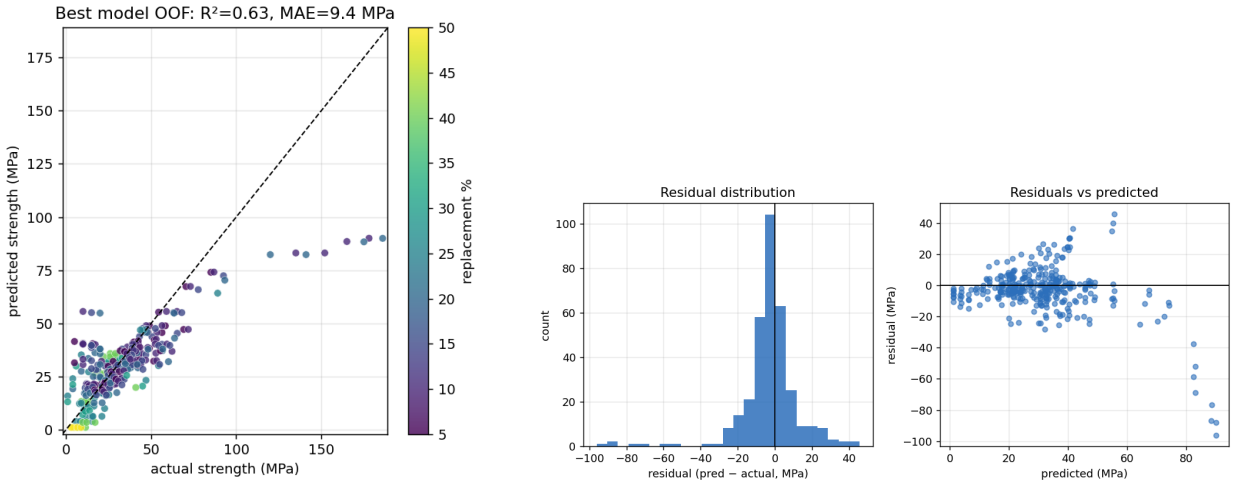
Feature	Meaning	Monotone constraint
`replacement_pct`	RHA cement-replacement level of the blend row (0–100%)	none
`age_days`	Curing age at which strength is measured (days)	non-decreasing
`ctrl_strength`	The *same paper's own* 0% (control) compressive strength at the matching curing age (MPa)	non-decreasing

The target is `strength_mpa`, the RHA-blend row's own compressive strength. `ctrl_strength` is looked up per `(paper_id, age_days)` from that paper's own 0%-replacement rows (Section 7 onward always assumes it is known at inference time, since it is the paper's own OPC baseline, not a property of the RHA blend being predicted). Combining these three features with monotone constraints and a light grouped hyper-parameter search yields a deliberately shallow model (`max_depth`=2, `learning_rate`=0.03, 300 trees). Its performance is

	R^2	RMSE (MPa)	MAE (MPa)
Final model (repeated grouped hold-out)	$\mathbf{0.50 \pm 0.20}$	**14.9**	**10.1**
Mean baseline	−0.31	24.0	17.4

Figure 4a shows out-of-fold predictions. Note that its panel title reports $R^2 = 0.63$, MAE = 9.4 MPa, which is *not* the headline number above: the scatter comes from a single 5-fold `GroupKFold` pass whose out-of-fold predictions are pooled into one R^2 over all 335 rows, whereas the table reports the mean of 40 grouped hold-outs that each train on fewer papers (70% vs 80%) and are scored split-by-split. The pooled single-pass figure is the more optimistic of the two; we quote the

repeated hold-out as the honest estimate and show the pooled predictions only because they give one prediction per row to plot. The model tracks normal-strength concrete well but systematically *under-predicts* the high end. Note that the 1–205 MPa span quoted in Study 1 describes framing A; framing C is evaluated on the 335 blend rows of 23 papers that also report a 0% control, and that subset reaches 186 MPa. Within it there is **exactly one** paper above 120 MPa (8 points, 120–186 MPa), and the model's predictions saturate near 90 MPa there, under-shooting by 70 MPa on average. The corpus does contain a second, even stronger paper—it holds the 205 MPa maximum—but it reports no control rows at all and is therefore excluded from framing C entirely, which is part of why 44 papers reduce to 23 here. With a single high-strength group in the evaluated set, grouped validation always places it wholly in the test fold, so the model has never seen a UHPC example when it is scored—an honest and interpretable failure mode, but one driven by a single group rather than by a subpopulation. That tail is also what pulls the residual distribution (Figure 4b) slightly negative: residuals are centred near zero (mean −2.7 MPa, median −2.2 MPa) but carry a long left tail reaching −100 MPa, all of it from that one paper. Feature importance (Figure 5) confirms that control strength (0.74) and curing age (0.17) carry most of the signal, with replacement level (0.09) modulating on top.



(a) Out-of-fold predictions.

(b) Residual diagnostics.

Figure 4: Final model behaviour, from a single pooled 5-fold `GroupKFold` pass ($R^2 = 0.63$ pooled, versus the 0.50 ± 0.20 repeated-hold-out estimate quoted in the table above). Colour encodes replacement level. The dashed line is the ideal $y = x$; the points falling far below it at the right of (a) are the single UHPC paper.

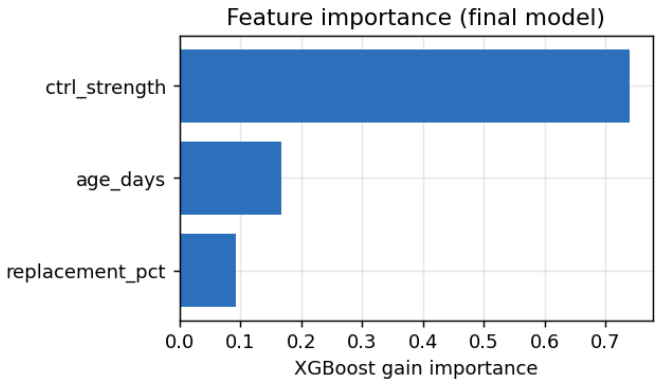


Figure 5: Feature importance of the final model. The paper's control strength and the curing age dominate; replacement level provides the RHA-specific modulation.

8 Predicting the Optimal Replacement Level

We derive the optimum as the replacement level that maximizes the predicted 28-day strength curve of the final model, evaluated on held-out papers, and also test a direct per-paper regressor. Neither beats simply predicting the corpus-mean optimum: on the 20 held-out papers with a reported optimum the argmax approach gives MAE = 4.75 percentage points against a mean-baseline of 3.45, and the direct regressor gives 6.5 against its own baseline of 5.3 over 43 papers. Figure 6 shows why, and the failure is starker than "fails to beat the baseline" suggests: **the argmax model returns the same value, 7.5%, for every one of the 21 held-out papers**. It is a constant predictor, and a miscalibrated one, since the corpus mean is 11.5%. This is a direct consequence of the final model's structure—it is monotone and non-decreasing in control strength and age, and with `max_depth`=2 there is almost no interaction between replacement level and the other two inputs, so changing a paper's control strength shifts the predicted strength curve vertically without moving the location of its maximum. Every paper therefore inherits the same argmax. The reported optima it is scored against are themselves concentrated but not tight: 75% of papers fall in 10–15% (mean 11.5%, median 10%), yet the full range runs 0–20% with a standard deviation of 4.6 points. A constant-11.5% prior already achieves MAE 3.45, so beating it requires resolving paper-to-paper shifts of a few percentage points—and with 20–43 labelled papers, and an interior optimum present in only 42% of them (Section 3), that signal is not available. Predicting the optimum is best treated as an open problem on this corpus rather than a solved output.

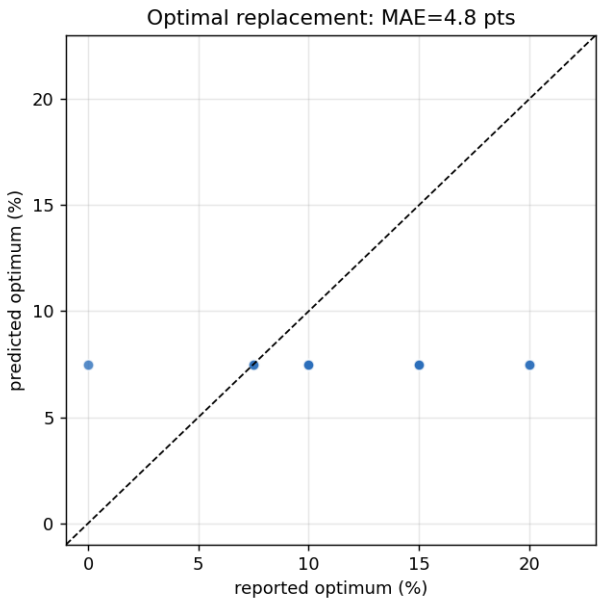


Figure 6: Predicted vs reported optimal replacement level. The predictions form a horizontal line: the model emits a constant 7.5% for every held-out paper, below the 11.5% corpus mean, and so loses to the mean prior it is compared against.

Part II — Sensitivity to the UHPC-Regime Outliers

Sections 7–8 flagged, but did not remove, the two papers responsible for the corpus's 1–205 MPa span. This part removes them and re-measures both target framings, to separate "the model degrades on UHPC" from "the model is fine once UHPC is taken out of scope."

9 Which Papers, and Why

The cutoff is applied at the *paper* level, not the row level: a paper is dropped in full—every replacement level, every curing age—if its **maximum** reported compressive strength exceeds **120 MPa**. This threshold is not a round-number guess; it sits in a genuine gap in the data. Of the 44 papers, only two ever cross it, and the next-highest paper after them tops out at exactly 100.0 MPa (Table 6).

ID	Title (abridged)	Max str.	Why it is out of scope
17	"...to produce ultra high performance concrete"	205.5 MPa	Explicitly UHPC. $w/b = 0.15$–0.23 vs. ≈ 0.35–0.50 for the rest of the corpus. Strength rises smoothly $51 \rightarrow 205.5$ MPa with age—internally consistent, just a different mix-design regime (silica fume / superplasticizer / very low w/b), not ordinary RHA concrete.
36	"Analysis of the effects of RHA on hydration..."	186.0 MPa	Reports 125–135 MPa at **1 day** of curing, including for its own 0% control—not physically plausible at any w/b. Every value is flagged `is_range=True` (read off a graph axis, not a table), consistent with a mis-read scale rather than genuine UHPC.

Table 6: The two papers removed by the 120 MPa cutoff, and the highest strength reported by any of the 42 papers kept: 100.0 MPa.

This directly tests the diagnosis already made in Section 7: that framing A's negative R^2 comes from mixing mortars, normal concrete and UHPC on one strength scale, and that framing C is largely insulated from this because it conditions on each paper's own control strength. Paper 17 was never in framing C to begin with (it reports no 0% control); paper 36 was one of its 23 papers.

Reproducibility check. Both sides below were re-run with the identical model class and hyperparameter grid as Sections 7–8. The unfiltered "before" run here reproduces the Section 7 headline almost exactly ($R^2 = 0.503 \pm 0.197$, RMSE= 14.90, MAE= 10.02 vs. the original $R^2 = 0.50 \pm 0.20$, RMSE= 14.9, MAE= 10.1), confirming the numbers below are directly comparable to the rest of this report.

10 Results With and Without the UHPC Papers

Framing A (absolute strength, no baseline info)	R^2	RMSE (MPa)	MAE (MPa)
With UHPC papers (44 papers, 756 rows, 1–205.5 MPa)	-0.43 ± 0.78	29.5	20.2
Without UHPC papers (42 papers, 706 rows, 1–100 MPa)	$\mathbf{+0.08 \pm 0.23}$	**17.1**	**13.0**

Framing C (tuned, blend | control)	R^2	RMSE (MPa)	MAE (MPa)	OOF R^2
With UHPC papers (23 papers, 335 rows)	0.50 ± 0.20	14.9	10.0	0.63
Without UHPC papers (22 papers, 327 rows)	$\mathbf{0.52 \pm 0.16}$	**11.4**	**8.6**	0.60

Optimal replacement (argmax, 28-day, held-out papers)	MAE (pts)	Mean-baseline (pts)
With UHPC papers (20 papers)	4.75	3.45
Without UHPC papers (20 papers)	4.75	3.45

11 Discussion

The two framings respond differently, and the difference is diagnostic rather than incidental. **Framing A flips sign**: R^2 crosses from -0.43 to $+0.08$, and RMSE/MAE fall by 42% and 36% respectively (29.5→17.1 MPa; 20.2→13.0 MPa). A model that is never told a paper is UHPC has no way

to know it should predict 180 MPa instead of 40 MPa for a nominally similar input row; once the two UHPC-regime papers are removed, the remaining 42 papers occupy one coherent strength scale (1–100 MPa) and the same features that were noise before now carry signal. $R^2 = 0.08$ is still weak in absolute terms—framing A was never going to become the recommended model—but the sign flip alone confirms the diagnosis. **Framing C improves modestly but genuinely**: R^2 rises from 0.50 to 0.52 with a visibly tighter spread across repeats ($\pm 0.20 \rightarrow \pm 0.16$), and RMSE/MAE both drop by double digits (14.9→11.4 MPa, −24%; 10.0→8.6 MPa, −14%). This is smaller than framing A's swing, exactly as Section 7 predicted: conditioning on the paper's own control strength already absorbs most of the cross-paper scale variation that framing A is exposed to raw, so removing the outlier papers here mostly trims a noisy, hard-to-fit paper (36) from a 23-paper set rather than unlocking new signal. One number moves the other way—the pooled single-pass OOF R^2 used for the prediction-scatter figure drops slightly (0.63 → 0.60)—which is expected: that figure is a single 5-fold split, more sensitive to exactly which papers land in which fold when there are only 22–23 of them, and is not the metric either number in the table above is drawn from. The optimal-replacement argmax MAE is *exactly* unchanged (4.75 points against an unchanged 3.45-point baseline, both sides): neither dropped paper had a usable reported optimum, so that evaluation set of 20 papers is identical before and after. The practical conclusion is that **outlier removal and the control-strength reframing are two different fixes for the same underlying problem**, and only one of them is needed if framing C is what will be deployed. Framing A's improvement after cleaning is useful mainly as confirmation that the corpus is well-behaved once UHPC is excluded, not as a replacement for framing C, which remains the stronger predictor either way (0.50–0.52 R^2 vs. 0.08 at best). If the eventual goal includes UHPC-range predictions, the two removed papers should be re-added as a separate, explicitly labelled regime (e.g. a categorical "mix class" feature) rather than silently dropped, since 8–38 data points is too little to train a UHPC-specific model on its own; for the conventional-concrete regime this report otherwise targets, dropping them is the right call.

12 An LLM Ablation: Does Control Strength Matter Across Model Families?

Sections 7 and Study 1 (Table 1) established, with a trained regressor, that supplying a paper's own 0% control strength is what makes RHA-blend strength learnable at all. Two independent LLM experiments ask the same question with no gradient-boosted model in the loop at all: given the same tabular rows, can a general-purpose LLM infer compressive strength from the paper's inputs alone, and does it too depend on being handed the control strength?

Setup. Six papers were held out at the *paper* level—P036, P022, P007, P030, P054 and P064, none seen in training—giving a fixed benchmark of **57 test points** shared identically by both conditions and both models, so the ablation isolates one variable:

WITH_CONTROL the model is given the paper's 28-day 0% control strength as an input feature, exactly as framing C does for the ML model.

NO_CONTROL control strength is withheld; only the RHA/mix inputs remain.

Two models. **Claude Fable** was evaluated externally (results as reported); **gpt-oss:120b-cloud** (OpenAI's open-weight ∼117B-parameter model, MXFP4-quantized, accessed through Ollama's cloud proxy) was run for this report directly against the identical ground truth, one Ollama

call per test paper per condition (12 calls total), each given that paper's held-out rows plus the *entire* condition's training CSV as in-context reference data, prompted to return a strict JSON array of predictions. gpt-oss was reached only after two earlier choices failed: DeepSeek-R1 8B run locally took 10+ minutes for a single call at this context length on an 8 GB laptop GPU (impractical for 12 calls), and `kimi-k2.6:cloud` returned 402 Payment Required (a paid ollama.com plan). gpt-oss:120b-cloud is free-tier accessible, cloud-hosted, and answered every call in 20–100 s. One call (NO_CONTROL, paper P007) initially degenerated—the model echoed the four *age* values back verbatim instead of predicting strength; it was retried once and produced a normal, non-degenerate answer, which is what is reported below. Because the two models were evaluated under different prompting protocols (Fable's exact prompt is not available to us), this is not a controlled head-to-head between Fable and gpt-oss—it is two *separate* within-model ablations that happen to share the same benchmark, and both are compared only against themselves (WITH_ vs NO_CONTROL), not against each other.

12.1 Overall Result

Condition	R^2	RMSE (MPa)	MAE (MPa)	Bias (MPa)
Fable, WITH_CONTROL	**0.877**	**8.79**	**5.84**	−2.21
Fable, NO_CONTROL	0.392	19.50	12.83	−6.25
gpt-oss:120b, WITH_CONTROL	**0.887**	**8.41**	7.17	−1.48
gpt-oss:120b, NO_CONTROL	0.487	17.92	12.78	−6.82
Final ML model (Section 7, for reference)	0.50 ± 0.20	14.9	10.1	—

Both LLMs move the same way. For Fable, control strength lifts R^2 by +0.485 and cuts RMSE/MAE by roughly **55%**. For gpt-oss, control strength lifts R^2 by +0.400 and cuts RMSE by **53%** (17.92→8.41 MPa) and MAE by **44%** (12.78→7.17 MPa)—the same qualitative effect Study 1 found for the ML model, now reproduced independently by *two* completely different LLMs. The ML row is included only as context, *not* as a matched comparison: it is a repeated grouped hold-out averaged over 40 different 70/30 paper splits, whereas all four LLM rows are single evaluations on one fixed six-paper split. Neither LLM's WITH_CONTROL number should be read as "beats the ML model"—the honest claim in both cases is the ablation *within* that model, which needs no such caveat because both conditions share the identical test papers.

12.2 Result by Compressive-Strength Category

Each test paper is assigned to one of three strength categories by its own 28-day control strength (Low < 30 MPa: P036, P022; Middle 30–60 MPa: P007, P030; High ≥ 60 MPa: P054, P064). The category is a property of the *paper*, not of each individual point—P054's points, for example, run from 42 to 78 MPa—and it splits the 57 test points 24/18/15. It reveals where control strength actually earns its keep, for both models.

	NO_CONTROL (MAE / Bias, MPa)				WITH_CONTROL (MAE / Bias, MPa)			
	Fable		gpt-oss		Fable		gpt-oss	
Regime (N)	MAE	Bias	MAE	Bias	MAE	Bias	MAE	Bias
Low, < 30 (24)	7.31	+7.31	7.02	+5.86	3.84	+3.70	5.59	+5.26
Mid, 30–60 (18)	3.19	−1.87	5.21	−3.51	3.85	−2.89	7.51	−4.14
High, ≥ 60 (15)	33.22	−33.22	31.09	−31.09	**11.43**	−10.85	**9.30**	−9.09

The pattern is sharp, matches the UHPC discussion of Section 9, and now holds for *both* LLMs independently: **without** control strength, both models are reasonably behaved in the low and

middle regimes but collapse on the high regime, under-predicting by a mean of 31–33 MPa—neither model has any way to know a given paper's own strength ceiling, so both regress toward the corpus-typical range. **With** control strength, high-regime MAE falls to 9.30–11.43 MPa (a ∼65–70% reduction) because the model is now told, explicitly, which papers are high-baseline. The low and middle regimes—already easy without any baseline signal—show smaller, noisier shifts (gpt-oss's mid-regime MAE actually rises slightly, 5.21 → 7.51, well within the noise of a 15–18-point evaluation). This mirrors exactly why framing C outperforms framing A for the XGBoost model: the hard part of this corpus was never the RHA effect itself, it is locating each paper's baseline strength level, and *any* sufficiently capable predictor—tree-based, or either of two unrelated LLM families—needs that baseline supplied to do well outside the middle of the strength range.

12.3 Where It Still Fails: Paper P064

Both models and both conditions agree on the single hardest case. Paper P064 (6 points, high replacement/high strength) is the worst-or-tied-worst per-paper error everywhere:

	MAE, NO_CONTROL		MAE, WITH_CONTROL	
Paper	Fable	gpt-oss	Fable	gpt-oss
P064 (hardest)	47.95	33.70	21.37	12.89
next-worst	23.40 (P054)	29.34 (P054)	4.81 (P054)	7.68 (P007)

Control strength shrinks P064's error substantially for both models (Fable 47.95 → 21.37; gpt-oss 33.70 → 12.89), but it remains the largest per-paper error by a wide margin in every configuration. At 28 days, NO_CONTROL predictions undershoot this paper's true 90–111 MPa range by 60–70 MPa (Fable) or land in the 40s-to-60s (gpt-oss)—even with age and replacement given, neither model can guess this paper's unusually high strength scale without being told it. This is the same failure mode already documented for the final ML model in Section 7 (systematic under-prediction at the high end, worst on the corpus's most extreme paper)—further evidence that the limitation is in the *data* (a handful of unusually strong papers with no comparable neighbors to generalize from), not specific to any one modeling approach.

12.4 Summary

Two unrelated LLMs, evaluated independently and under different prompting protocols, replicate this report's central empirical claim, first established with a three-feature, monotone, depth-2 XGBoost tree: **a paper's own control strength is the dominant signal for predicting RHA-blend compressive strength, and its absence is felt almost entirely at the high-strength end of the corpus.** That three completely different predictors—a gradient-boosted tree and two general-purpose LLMs from different vendors—land on the same qualitative conclusion, on the same held-out papers, is considerably stronger evidence for that claim than any single result alone.

Part III — Predicting Strength up to 60 MPa, Given the Control Strength

Part III narrows the problem to the case of practical interest:

- **Only concrete up to 60 MPa.** Every data point with compressive strength above 60 MPa is removed; the high-strength regime is not of interest.
- **The control strength is always an input** (framing C). Alongside the RHA replacement level and curing age, the model is told the strength of the same paper's plain concrete (0% RHA). This is known in practice—the plain mix can always be tested first—and Parts I and II showed it is the most useful input.

Within this one setting, Part III runs an ablation study for the machine-learning model (Section 14), makes gpt-oss as reproducible as possible (Section 15), and compares the two on identical test points (Sections 16 and 17).

13 Data

Removing the points above 60 MPa leaves the following data.

Dataset	Before	After (≤ 60 MPa)
ML corpus: RHA-blend rows with a control, papers	335 rows, 23 papers	**308 rows, 21 papers**
LLM benchmark: training rows	215	**193**
LLM benchmark: test points (6 held-out papers)	57	**49**

In every ML experiment, each row is one RHA blend *(paper, replacement %, curing age)*; its inputs are the replacement level, the age and the paper's control strength at the same age, and the target is its measured strength.

Same evaluation as Part I. The ML results use exactly the evaluation of Part I: the same scripts (`06_ablation.py`, `07_best_model.py`) and the same 40 repeated grouped hold-outs (30% of papers held out each time, so no paper is in both training and test), reported as mean ± standard deviation. The only difference between the Part I and ≤ 60 MPa columns below is the data.

How to read the comparison. RMSE and MAE are errors in MPa and can be compared directly. R^2 cannot, quite: it measures error relative to the spread of the true strengths, and removing the points above 60 MPa almost halves that spread (standard deviation 25.9 MPa $\rightarrow$ 13.7 MPa). So a model can have smaller errors on the ≤ 60 MPa data and still a lower R^2.

14 Machine-Learning Approach: Ablation Study

Four studies each vary one design choice and keep the rest fixed (Table 7). The best value of each study is in bold, separately for Part I and for ≤ 60 MPa.
The feature groups are the same as in Part I: *base* is replacement level, age and control strength; *chemistry* the oxide contents, amorphous content and loss on ignition; *fineness* particle size, surface area and specific gravity; *processing* burning temperature and duration, grinding time and w/b ratio; *categorical* activation method, mineralogy, processing type and pozzolanic reactivity; *text* an 8-number summary of each paper's free-text description. *Monotone* forces predicted strength never to fall as age or control strength rise.

Table 7: Ablation study with the control strength as input: Part I (all points) against Part III (points ≤ 60 MPa), same 40 repeated grouped hold-outs.

		Part I (all points)			Part III (≤ 60 MPa)		
Study	Configuration	R^2	RMSE	MAE	R^2	RMSE	MAE
1 Model (all features)	Random Forest	**0.436 ± 0.215**	**15.82**	**10.40**	**0.202 ± 0.299**	**11.97**	**8.83**
	XGBoost	0.395 ± 0.259	16.45	10.64	0.197 ± 0.254	12.08	8.95
	HistGradientBoosting*	−0.524 ± 0.710	22.28	15.83	0.079 ± 0.304	12.55	9.07
	Mean baseline	−0.310 ± 0.366	23.97	17.40	−0.147 ± 0.267	14.48	12.32
	Ridge regression	−14.7 ± 46.1	44.74	27.65	−11.2 ± 32.4	31.89	20.25
2 Features (XGBoost)	base [replacement, age, control]	**0.442 ± 0.213**	**15.54**	10.66	0.185 ± 0.273	12.17	9.04
	base + chemistry	0.317 ± 0.355	17.18	11.24	0.111 ± 0.289	12.67	9.47
	base + fineness	0.435 ± 0.297	15.73	**10.50**	0.210 ± 0.237	12.01	8.96
	base + processing	0.236 ± 0.584	17.69	12.34	0.193 ± 0.236	12.16	9.58
	base + categorical	0.427 ± 0.221	15.93	10.86	0.243 ± 0.246	11.71	9.02
	base + all numeric	0.349 ± 0.336	16.85	11.01	0.197 ± 0.277	12.05	8.89
	all structured	0.395 ± 0.259	16.45	10.64	0.197 ± 0.254	12.08	8.95
	all + TF-IDF text	0.403 ± 0.238	16.42	10.72	0.192 ± 0.269	12.12	8.97
	all + MiniLM text	0.426 ± 0.215	16.13	10.53	**0.265 ± 0.262**	**11.54**	**8.57**
3 Missing values (XGBoost)	native (no imputation)	**0.397 ± 0.246**	16.40	10.60	0.207 ± 0.273	11.98	8.79
	median imputation*	0.396 ± 0.221	**14.88**	**9.74**	0.191 ± 0.245	11.81	**8.59**
	KNN imputation*	0.390 ± 0.268	14.92	10.14	**0.218 ± 0.250**	**11.57**	8.77
	MICE imputation*	0.249 ± 0.322	16.35	11.17	0.083 ± 0.239	12.60	9.60
4 Physics priors (XGBoost, base)	plain	0.442 ± 0.213	15.54	10.66	0.185 ± 0.273	12.17	9.04
	log-target	0.423 ± 0.178	16.09	10.87	0.080 ± 0.304	12.92	9.52
	monotone in age and control	**0.482 ± 0.217**	**15.02**	**10.21**	**0.308 ± 0.262**	**11.16**	**8.52**
	log-target + monotone	0.458 ± 0.196	15.71	10.63	0.203 ± 0.300	11.97	9.04

*Fewer than 40 splits could be fitted: 31 in Part I, 21 for ≤ 60 MPa. Errors are in MPa.

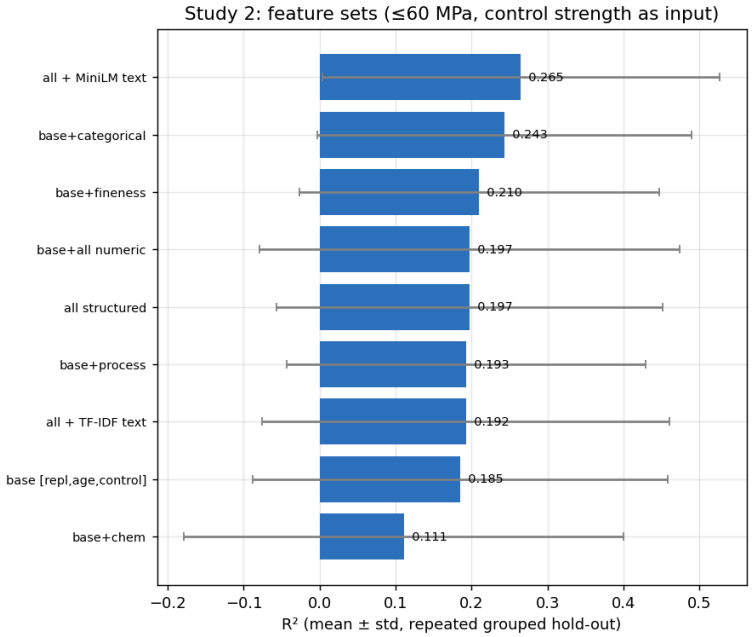


Figure 7: Study 2 on the ≤ 60 MPa data: every feature set lies within one standard deviation of every other (compare Figure 3 for Part I).

What the comparison shows.

- **Errors are smaller on the ≤ 60 MPa data in every row.** For the XGBoost and Random Forest configurations, RMSE drops by 3–5.5 MPa (for example 16.45 → 12.08 MPa for XGBoost with all features) and MAE by 1–3 MPa. R^2 is lower for all of these, for the reason given above; only the weak models (HistGradientBoosting, mean baseline, Ridge) score higher R^2 on the ≤ 60 MPa data.
- **Model:** in both datasets Random Forest and XGBoost are the two best models and are within noise of each other; Ridge fails in both. HistGradientBoosting recovers from failing badly (R^2 −0.52) to a positive score once the extreme points are gone.
- **Features:** in Part I the three base inputs were best and most added groups made things worse. On the ≤ 60 MPa data the order changes—text embeddings (0.265) and categorical descriptors (0.243) score highest, the base set 0.185—but all sets are within one standard deviation of each other, so **no feature group reliably adds anything beyond replacement, age and control strength** in either dataset. Chemistry is among the worst in both.
- **Missing values:** in both datasets native handling, median and KNN imputation are equivalent and MICE is worst.
- **Physics priors:** the monotone constraint is the best choice in both datasets, and its gain is larger on the ≤ 60 MPa data (+0.12 R^2, against +0.04 in Part I).

The winning recipe is therefore the same in both: XGBoost, the three base inputs, and monotone constraints.

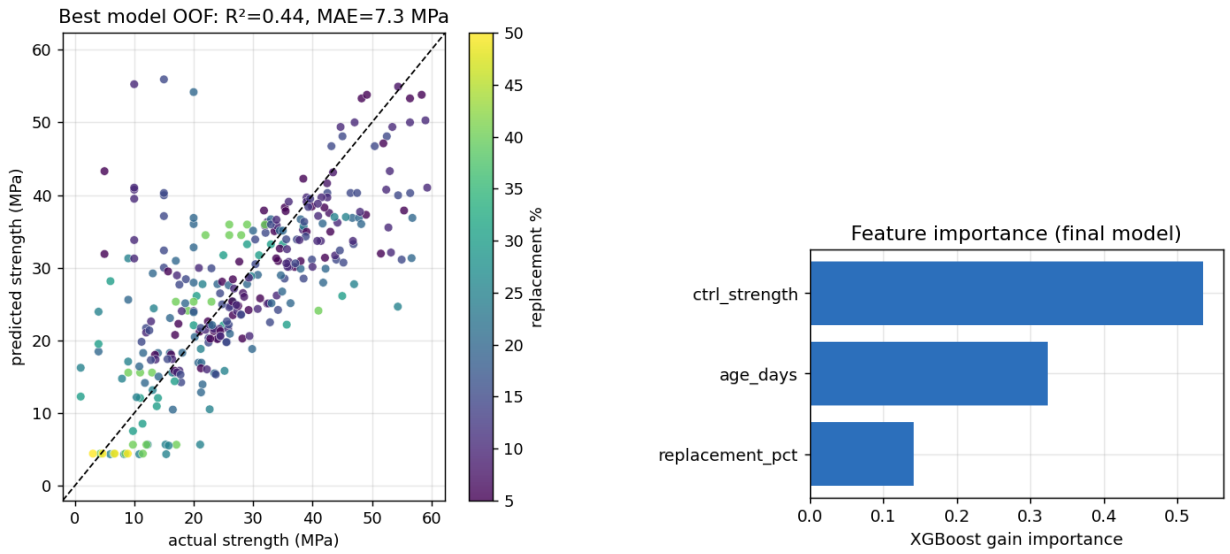
14.1 Final Model

That recipe, with the XGBoost settings tuned by grouped grid search on each dataset:

	Part I (all points)	Part III (≤ 60 MPa)
Blend rows / papers	335 / 23	308 / 21
Strength range (spread, SD)	1–186 MPa (25.9)	1–59 MPa (13.7)
R^2 (40 repeated grouped hold-outs)	$\mathbf{0.50 \pm 0.20}$	0.35 ± 0.25
RMSE (MPa)	14.9	**10.9**
MAE (MPa)	10.1	**8.2**
Pooled 5-fold grouped R^2 (scatter plots)	0.63	0.44
Worst out-of-fold under-prediction	−100 MPa	**−30** MPa
Tuned settings	depth 2, min child 3, $\lambda = 1$	depth 4, min child 5, $\lambda = 3$
Importance: control / age / replacement	0.74 / 0.17 / 0.09	0.54 / 0.32 / 0.14
Optimum-% MAE (average-optimum baseline)	4.75 (3.45) pts	4.17 (2.87) pts

On the ≤ 60 MPa data the final model's RMSE is 27% smaller (14.9 → 10.9 MPa) and its MAE 19% smaller (10.1 → 8.2 MPa), and the large under-predictions of Part I, which all came from one UHPC paper, are gone (worst miss 30 MPa instead of 100 MPa; Figure 8). R^2 is lower (0.50 → 0.35) because the target spread is halved, not because the model is worse. Using Part I's tuned settings unchanged on the ≤ 60 MPa data gives the same result (0.35 ± 0.27, RMSE 10.4, MAE 8.0 MPa), so the settings transfer. With the extreme papers removed, age and replacement level carry more of the signal than before. In both datasets the model predicts the same optimum replacement for

every held-out paper (7.5%) and loses to predicting the average optimum, for the reason explained in Section 8.



(a) Out-of-fold predictions.

(b) Feature importance.

Figure 8: Final model on the ≤ 60 MPa data (single pooled 5-fold grouped pass, pooled R^2 0.44). Compare Figures 4a and 5 for Part I.

15 LLM Approach: gpt-oss with Maximum Reproducibility

gpt-oss:120b-cloud (run through Ollama's cloud service) is given the 193 training rows in its prompt and asked to predict the 49 test points of the six held-out papers. Each test row includes the paper's 28-day control strength.

15.1 One Fixed Prompt

Every call uses the same versioned prompt template, `pipeline/prompts/rha_strength_v1.txt`, filled in the same way each time (training rows and test queries sorted by paper, replacement level and age). The finished prompt was byte-identical on all calls (same SHA-256 hash) and the model version was identical on all calls (Ollama digest `ac7f7a1e7785`), so any difference between answers comes from the model service, not from anything we changed.

```
You are a concrete materials scientist predicting the compressive strength of
rice-husk-ash (RHA) blended concrete and mortar from published literature data.
All strengths in this task are 60 MPa or below.

TRAINING DATA
Below are {n_train} measured data points from other papers, as CSV with a header
row. Each row is one (paper, RHA replacement %, curing age) combination with its
measured compressive_strength_mpa. Learn how the input columns relate to
compressive_strength_mpa.
--- BEGIN TRAINING CSV ---
{train_csv}
--- END TRAINING CSV ---

TASK
Predict compressive_strength_mpa for the {n_queries} query rows below. They come
from papers that are NOT in the training data. Each row states its paper_id; rows
with the same paper_id share one mix design, rows with different paper_ids do not.
Each query includes control_strength_28d_mpa: the SAME paper's own 0% RHA (plain
cement) 28-day compressive strength. Use it as that paper's baseline strength level.

{queries}

OUTPUT FORMAT
Return only a JSON array of exactly {n_queries} numbers: one predicted compressive
strength in MPa per query, in query order (Query 1 first). No units, no text, no
code fence. Example for three queries: [12.3, 45.6, 30.1]
```

15.2 Ablation of Reproducibility Settings

Settings were added one at a time, and the prompt was sent five times under each:

S0 temperature 0 and a fixed seed.

S1 + fully greedy decoding (always take the single most likely token: `top_k` = 1, `top_p` = 1, no repetition penalty) and a fixed output length cap.

S2 + a JSON schema requesting exactly 49 numbers. (The cloud service accepted this option but did not enforce it.)

S3 + low reasoning effort, so the model thinks for less time before answering.

S4 one short prompt per paper instead of one prompt for all six papers.

	Single run (5 runs)					Median of
Setting	R^2	R^2 range	RMSE	MAE	Query SD	5 runs, R^2
S0 temperature 0 + seed	0.407 ± 0.293	-0.05 to 0.77	12.39	8.87	5.4	0.587
S1 + greedy decoding	$\mathbf{0.585 \pm 0.154}$	0.31 to 0.76	**10.53**	**7.82**	**3.1**	0.711
S2 + JSON schema	0.465 ± 0.402	-0.33 to 0.77	11.49	8.61	6.1	0.769
S3 + low reasoning	0.418 ± 0.395	-0.23 to 0.83	11.93	9.19	6.8	0.711
S4 one prompt per paper	0.552 ± 0.134	0.36 to 0.71	10.99	7.93	4.1	0.647

Query SD is how much a single prediction moves between runs (median over the 49 test points, in MPa). The last column scores the per-point median of the five runs.

- **No setting makes gpt-oss reproducible.** Under every setting, all five runs gave different answers, and no test point got the same prediction in all five. Temperature 0 and greedy

decoding fix how the next word is *chosen*, but the probabilities it is chosen from are computed on shared cloud GPUs, where tiny rounding differences depend on which other users' requests are processed at the same time. Once one token differs, the rest of the answer differs.

- **Greedy decoding (S1) gave the smallest run-to-run spread and the best accuracy**, so it was used for the final runs. The other settings did not reduce the spread; low reasoning effort only halved the time per call (12 vs 24–29 s).
- **Taking the median of five runs helps under every setting**: R^2 0.59–0.77, above the average single run in each case.

15.3 Final Runs

The S1 setting was run 15 times. Two answers were unusable (one had 50 numbers instead of 49, one was empty) and are excluded; the 13 valid runs were combined in two independent groups of five by taking the median prediction for each test point.

	R^2	RMSE (MPa)	MAE (MPa)	Run-to-run disagreement
Single run (13 valid; R^2 from -0.72 to 0.77)	0.486 ± 0.376	11.39	8.57	7.1 MPa per point
Median of 5 runs, group 1	**0.711**	**8.93**	**6.68**	
Median of 5 runs, group 2	**0.695**	**9.18**	**7.34**	1.0 MPa per point between groups

A single gpt-oss run can land almost anywhere, from R^2 -0.72 to 0.77. The median of five runs is both more accurate ($R^2 \approx 0.70$) and consistent: two completely independent groups of five agree to within about 1 MPa per test point.

15.4 Recommended Production Recipe

1. Pin the model version, and refuse to run if it changes.
2. Use one versioned prompt template, filled in deterministically.
3. Decode greedily: temperature 0, fixed seed, `top_k` = 1, `top_p` = 1, fixed length cap.
4. Check every answer (exactly one number per test point, all within 0–60 MPa) and retry the ones that fail: 2 of 15 final calls would have needed a retry.
5. Run five valid answers and use the per-point median.
6. Cache the final answer, keyed by prompt and model version, so the same request always returns exactly the same result.

16 Machine Learning vs gpt-oss on the Same 49 Points

For a fair comparison, the final XGBoost model was trained on the LLM benchmark's own 193 training rows, using replacement level, age and the 28-day control strength (the same information gpt-oss receives), and scored on the same 49 test points.

Model	R^2	RMSE (MPa)	MAE (MPa)	Bias (MPa)	Same answer every time?
XGBoost (final model)	**0.778**	**7.82**	**6.21**	-3.60	yes (identical)
gpt-oss, median of 5 runs	0.703	9.06	7.01	-2.72	within ≈ 1 MPa
gpt-oss, single run	0.486 ± 0.376	11.39	8.57	-1.79	no

By strength category (each test paper's category is set by its control strength, as in Section 12.2; MAE in MPa):

Category (test points)	XGBoost	gpt-oss, median of 5	gpt-oss bias
Low, control < 30 MPa (24)	**4.1**	4.4	+4.2
Middle, control 30–60 MPa (17)	7.5	**7.3**	−7.2
High-baseline papers, control ≥ 60 MPa (8)	**10.0**	14.1	−14.1

The three-input XGBoost model is more accurate than gpt-oss (R^2 0.78 vs 0.70), gives exactly the same answer every time, and runs in milliseconds instead of a 30-second cloud call. The two are level on low- and middle-category papers; XGBoost's advantage comes from the two papers with the strongest control concrete, which gpt-oss under-predicts by 14 MPa on average. With only six test papers, a gap of this size is suggestive rather than conclusive.

17 gpt-oss on the Same 40 Grouped Hold-Outs as XGBoost

Section 16 used a single split of six papers. Here gpt-oss is evaluated exactly as the ML models were: on the same 40 repeated grouped hold-outs of the same 308 blend rows from 21 papers. Each split holds out 7 papers (78–141 test points); the remaining ≈209 rows go into the prompt as training data. The held-out papers were checked to be identical to XGBoost's in all 40 splits.

- **Inputs:** every available feature—replacement level, age, the same-age control strength, and the chemistry, fineness, processing and categorical descriptors (the XGBoost "all features" set).
- **Prompt and settings:** the template of Section 15, with the control sentence describing the same-age control, and the greedy settings (S1).
- **Runs:** five per split (200 calls), each answer checked and retried up to twice if invalid. 180 answers were valid first time and 20 needed a retry; all 200 were valid in the end. Each call took 33 s on average (1.8 h in total).

Model (40 grouped hold-outs, ≤ 60 MPa)	R^2 (mean ± SD)	Median R^2	RMSE (MPa)	MAE (MPa)
XGBoost, final model (3 inputs, monotone)	$\mathbf{0.346 \pm 0.249}$	**0.450**	**10.86**	8.18
XGBoost, all features	0.197 ± 0.254	0.229	12.08	8.95
gpt-oss, single run	0.144 ± 0.597	0.294	11.94	8.55
gpt-oss, median of 5 runs	0.259 ± 0.518	0.444	11.10	**7.85**

Because both models were scored on the same 40 splits, they can also be compared split by split (gpt-oss median of 5 runs against each XGBoost model; p from a Wilcoxon signed-rank test on the 40 paired values):

gpt-oss (median of 5) against	Higher R^2 in	Lower MAE in	p for MAE
XGBoost, final model	19 of 40 splits	27 of 40 splits	0.024
XGBoost, all features	29 of 40 splits	32 of 40 splits	0.001

What this shows.

- **On a typical split, gpt-oss matches the best XGBoost model.** The median R^2 is the same (0.444 vs 0.450), and gpt-oss's MAE is slightly but significantly lower (7.85 vs 8.18 MPa, $p = 0.02$). The differences in RMSE and R^2 are not significant ($p = 0.87$ and 0.94).
- **Given the same information, gpt-oss beats XGBoost.** Against XGBoost with all features it has lower MAE in 32 of 40 splits ($p = 0.001$). XGBoost is only better when it is limited to the three base inputs with monotone constraints.
- **gpt-oss fails badly on a few splits**, which lowers its mean R^2 (0.26 vs 0.35) and doubles its spread (± 0.52 vs ± 0.25). On the two worst splits it scores R^2 -1.44 and -1.25, where XGBoost scores -0.03 and 0.15. In both, gpt-oss over-predicts corpus papers 22, 27 and 78 by about 20 MPa (the ML corpus numbers its papers differently from the six-paper benchmark). In these papers RHA cuts strength to 49–70% of the control on average, against 91% for a typical paper; gpt-oss assumes the blend stays close to its control strength, while XGBoost has learned from the training papers that high replacement lowers strength.
- **Single runs are clearly worse**: R^2 0.144 ± 0.597. Re-running the whole 40-split evaluation gives a different mean each time (0.11 to 0.17 over five passes), and one prediction moves by 3.5 MPa between runs (median). Taking the median of five runs raises the mean R^2 by 0.12, so the production recipe of Section 15.4 matters.
- **Cost:** five 33-second cloud calls per split for gpt-oss, against milliseconds for XGBoost, which also gives the same answer every time.

In short, gpt-oss is as accurate as the best XGBoost model on typical papers, but it breaks down on papers where RHA sharply lowers strength. The XGBoost final model is the more reliable choice.

18 Part III Summary

- **Setting:** concrete up to 60 MPa, with the paper's control strength as an input (308 rows from 21 papers for ML; 49 test points from 6 papers for the LLM comparison).
- **Best ML model, same evaluation as Part I:** XGBoost on replacement level, age and control strength, with strength forced to rise with age and control strength. Against Part I, its errors are smaller (RMSE 14.9 $\rightarrow$ 10.9 MPa, MAE 10.1 $\rightarrow$ 8.2 MPa) and its worst miss drops from 100 to 30 MPa; its R^2 is lower (0.50 $\rightarrow$ 0.35) only because the ≤ 60 MPa data has half the spread.
- **Ablation:** the same recipe wins on both datasets. The monotone constraint is the most valuable choice; no extra feature group reliably improves on the three base inputs; the model type and the way missing values are handled make little difference.
- **gpt-oss is not reproducible** under any decoding setting, even at temperature 0. Taking the median of five runs makes it accurate (R^2 0.70) and consistent (1 MPa agreement); caching makes it identical.
- **On the same 49 test points, XGBoost beats gpt-oss** (R^2 0.78 vs 0.70) and is fully reproducible, fast and free to run.
- **On XGBoost's own 40 grouped hold-outs** (all features, median of five runs), gpt-oss matches the best XGBoost model on a typical split (median R^2 0.44 vs 0.45; MAE 7.85 vs

8.18 MPa) but fails badly on papers where RHA sharply lowers strength, so its mean R^2 is lower (0.26 vs 0.35) and twice as variable. XGBoost remains the more reliable choice.